

# **CanSat Avionics Wiring Documentation**

## **Telemetry Architecture**

**Subsystem:** Avionics

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# Overview

This documentation describes the avionics wiring for both the onboard and ground station setups. The primary topics include the component lists, important design considerations, and online schematics.

## Onboard Flight Computer Component List

| Component                          | Function                                                            |
|------------------------------------|---------------------------------------------------------------------|
| Arduino Uno / Nano                 | Flight computer, executes mission logic                             |
| 2 × 18650 Li-ion battery pack      | Primary power source (6.0–8.4 V unregulated)                        |
| 5 V buck regulator                 | Powers Arduino, servo, and 5 V-tolerant sensors                     |
| 3.3 V regulator / NRF24L01 adapter | Powers the radio module                                             |
| BMP280                             | Barometric pressure sensor, used for altitude and descent detection |
| MPU6050                            | Inertial measurement unit (acceleration and rotation rate)          |
| NEO-6M GPS                         | Position tracking (latitude/longitude)                              |
| NRF24L01                           | 2.4 GHz wireless telemetry radio                                    |
| SG90 servo                         | Actuates the release mechanism                                      |

### Schematic files:

- Onboard flight computer:  
<https://app.cirkitdesigner.com/project/68f9033c-53bc-4e92-ae24-991168581010>
- Ground station:  
<https://app.cirkitdesigner.com/project/a2922b54-466c-49a2-95f8-4cfd1af1ee90>

## Onboard Computer Power Distribution

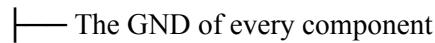
The given power source includes a 2 × 18650 battery pack. This battery pack has a raw voltage of 6.0-8.4 V. This voltage exceeds the more common 3.3 V- 5 V range that nearly all of our components operate under. Thus, the power flow of the system must be designed to account for this disparity. The power rail logic revolves around power moving from the battery, towards a switch, and then to voltage regulators

that will step the voltage down to usable values. Only after this step can any power reach the avionics. This is clearly outlined in the online schematics for both onboard and ground station computers. Below is a rough visualization in case the schematic is unclear.

Battery (+) → Accessible Power Switch



Battery (−) → Common Ground Node



The 5 V regulator supplies only components that require 5 volts, including the Arduino and servo. It is important to note that the EN and Vin pins are both connected to power. This ensures that the regulator is consistently regulating voltage during the entire mission.

In the case of the 3.3V regulator, the relevant connected components include the NRF, GPS, MPU, and BMP modules.

It is good practice to use capacitors to regulate power and ensure the system operates consistently under stress. In this circuit, the capacitors are placed in parallel and not in series at two locations.

| Location              | Purpose                                         |
|-----------------------|-------------------------------------------------|
| NRF24L01 power pins   | Stabilizes radio during transmit/receive spikes |
| 5 V rail/servo output | Absorbs current draw during servo usage         |

## Onboard Computer Wiring Reference

| Component | Pin    | Arduino Pin / Rail | Notes                                     |
|-----------|--------|--------------------|-------------------------------------------|
| Arduino   | 5V     | 5 V regulator OUT+ | Regulated supply only                     |
| Arduino   | GND    | Common GND         | —                                         |
| Servo     | Signal | D3                 | Release actuation control                 |
| Servo     | VCC    | 5 V rail           | Do not power from Arduino 5V pin directly |
| Servo     | GND    | Common GND         | —                                         |

| Component  | Pin  | Arduino Pin / Rail   | Notes                               |
|------------|------|----------------------|-------------------------------------|
| BMP280     | SDA  | A4                   | I2C data                            |
| BMP280     | SCL  | A5                   | I2C clock                           |
| BMP280     | VCC  | 5 V or 3.3 V         | Verify breakout board rating        |
| BMP280     | GND  | Common GND           | —                                   |
| MPU6050    | SDA  | A4                   | Shared I2C bus with BMP280          |
| MPU6050    | SCL  | A5                   | Shared I2C bus with BMP280          |
| MPU6050    | VCC  | 5 V or 3.3 V         | Verify breakout board rating        |
| MPU6050    | GND  | Common GND           | —                                   |
| NEO-6M GPS | TX   | D4                   | GPS → Arduino data line             |
| NEO-6M GPS | RX   | Disconnected (or D5) | Only required for GPS configuration |
| NEO-6M GPS | VCC  | 5 V rail             | Verify breakout board rating        |
| NEO-6M GPS | GND  | Common GND           | —                                   |
| NRF24L01   | CE   | D9                   | Radio enable                        |
| NRF24L01   | CSN  | D10                  | SPI chip select                     |
| NRF24L01   | MOSI | D11                  | Arduino → radio                     |
| NRF24L01   | MISO | D12                  | Radio → Arduino                     |
| NRF24L01   | SCK  | D13                  | SPI clock                           |
| NRF24L01   | VCC  | 3.3 V only           | Never connect to 5 V                |
| NRF24L01   | GND  | Common GND           | —                                   |

| Component | Pin | Arduino Pin / Rail | Notes                              |
|-----------|-----|--------------------|------------------------------------|
| NRF24L01  | IRQ | Not connected      | Reserved, unused in current design |

**Note on shared I2C bus:** The BMP280 and MPU6050 share the A4 (SDA) and A5 (SCL) lines, relying on distinct I2C addresses to avoid bus conflicts. It is also important to note that some connections might have been missed in the table, so refer to the online circuit diagrams for an accurate representation.

## Ground Station Telemetry

The onboard radio communicated with the ground station radio to transmit telemetry data. As such, the ground station uses the same radio module that operates at 2.4GHz and communicates with the Arduino via SPI. The ground station primarily has the same wiring as the onboard flight computer, but with only the radio, 3.3V regulator, Arduino, and capacitor. The Arduino will be connected via USB to a computer to display live telemetry results and to log everything.

**Required:** Laptop, USB cable, second Arduino, second NRF24L01, 3.3 V regulator, 100  $\mu$ F capacitor

**Not required:** Environmental/motion sensors, GPS, servo, or CanSat battery (the ground Arduino is powered via USB)

The power rail principle presented for the onboard flight computer carries over to the ground station computer. In this case, there is only the 3.3V regulator, and no need for a 5V regulator. To ensure consistent power throughout the system, a capacitor is present to ensure no dips or spikes in power while receiving from the CanSat.

## Ground Station Computer Wiring Reference

| Component | Pin  | Ground Arduino Pin / Rail      | Notes           |
|-----------|------|--------------------------------|-----------------|
| NRF24L01  | VCC  | 3.3 V regulator/adapter output | Radio power     |
| NRF24L01  | GND  | Common GND                     | —               |
| NRF24L01  | CE   | D9                             | Radio enable    |
| NRF24L01  | CSN  | D10                            | SPI chip select |
| NRF24L01  | MOSI | D11                            | Arduino → radio |

| Component | Pin  | Ground Arduino Pin / Rail | Notes                              |
|-----------|------|---------------------------|------------------------------------|
| NRF24L01  | MISO | D12                       | Radio → Arduino                    |
| NRF24L01  | SCK  | D13                       | SPI clock                          |
| NRF24L01  | IRQ  | Not connected             | Reserved, unused in current design |

Because the PDR required a 2.4 GHz antenna/radio module operating at 3.3 V over a wired SPI connection to the flight computer, this specific setup was chosen. The display should handle the following:

- CSV logging of all received telemetry
- Live numeric display of current values
- Altitude-vs-time plotting
- Rotation-rate-vs-time plotting
- GUI display of the above in real time

## Operational Sequence

| Phase                     | Description                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Power-on / pre-launch  | Regulators energize; Arduino initializes all sensors and sets the servo to a locked position                                               |
| 2. Ascent                 | BMP280 continuously measures pressure; Arduino converts readings to altitude and monitors for the release condition                        |
| 3. Release command        | Arduino commands the servo (D3) to actuate the release latch at the target altitude                                                        |
| 4. Descent detection      | Arduino compares the three most recent altitude readings; a decrease of $\geq 1$ m confirms descent                                        |
| 5. Telemetry transmission | Arduino assembles a packet (sequence number, altitude, GPS position, rotation rate) and transmits it via the NRF24L01 at least three times |

| Phase               | Description                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| 6. Landing/recovery | CanSat is recovered, powered off via the accessible switch, and telemetry is reviewed to verify mission success |

## Assembly and Verification Procedure

To reduce the risk of damaging components:

1. Wire only the battery, switch, and regulators; leave all logic components disconnected
  2. Power on and use a multimeter to confirm the 5 V rail reads approximately 5.0 V and the 3.3 V rail reads approximately 3.3 V
  3. Power off, then connect the Arduino; power on and confirm normal boot behaviour
  4. Power off, then connect sensors (BMP280, MPU6050, GPS)
  5. Power off, then connect the NRF24L01
  6. Connect the servo last, as it is the highest-current, highest-risk component
- The NRF24L01 must never be powered from the 5 V rail
  - Raw battery voltage must never be connected directly to the Arduino 5V pin
  - All components must share a common ground reference

## Design Summary

This architecture reflects three core electrical design principles applied throughout the subsystem:

1. **Voltage isolation:** no component is exposed to unregulated battery voltage
2. **Common ground referencing:** a single ground ties all subsystems together
3. **Protection:** Capacitors are placed at both current-sensitive points (radio) and high-draw points (servo) to maintain rail stability during operation